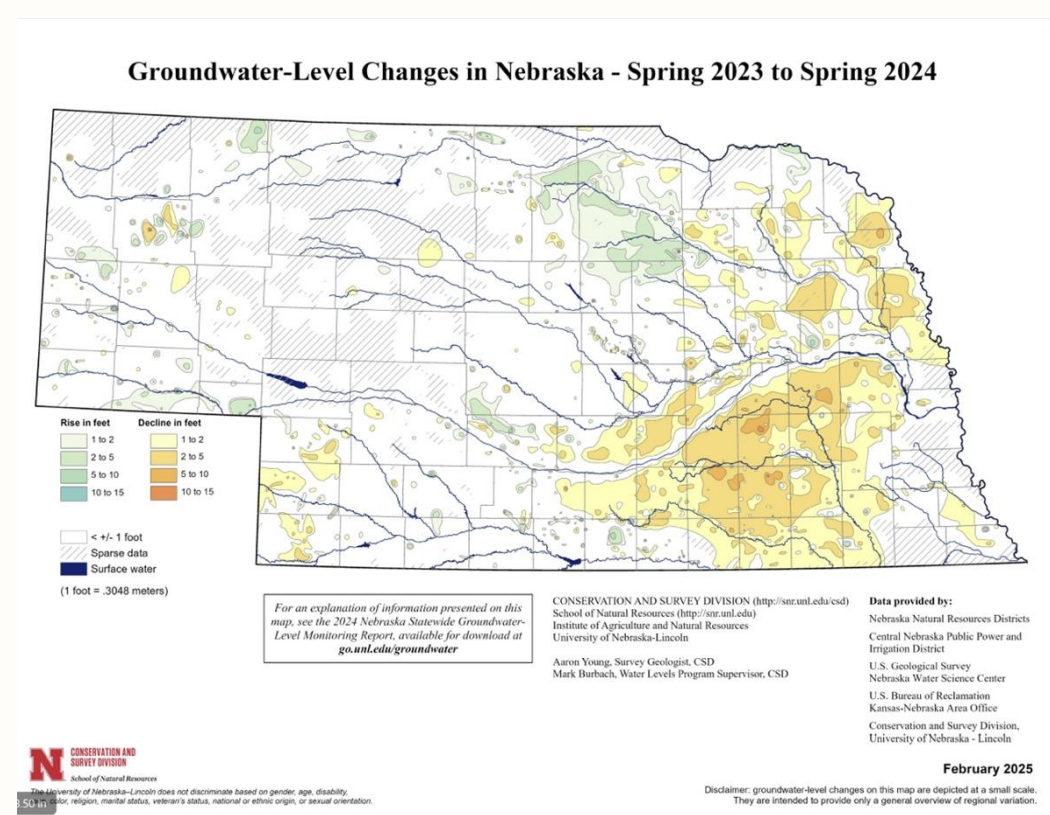
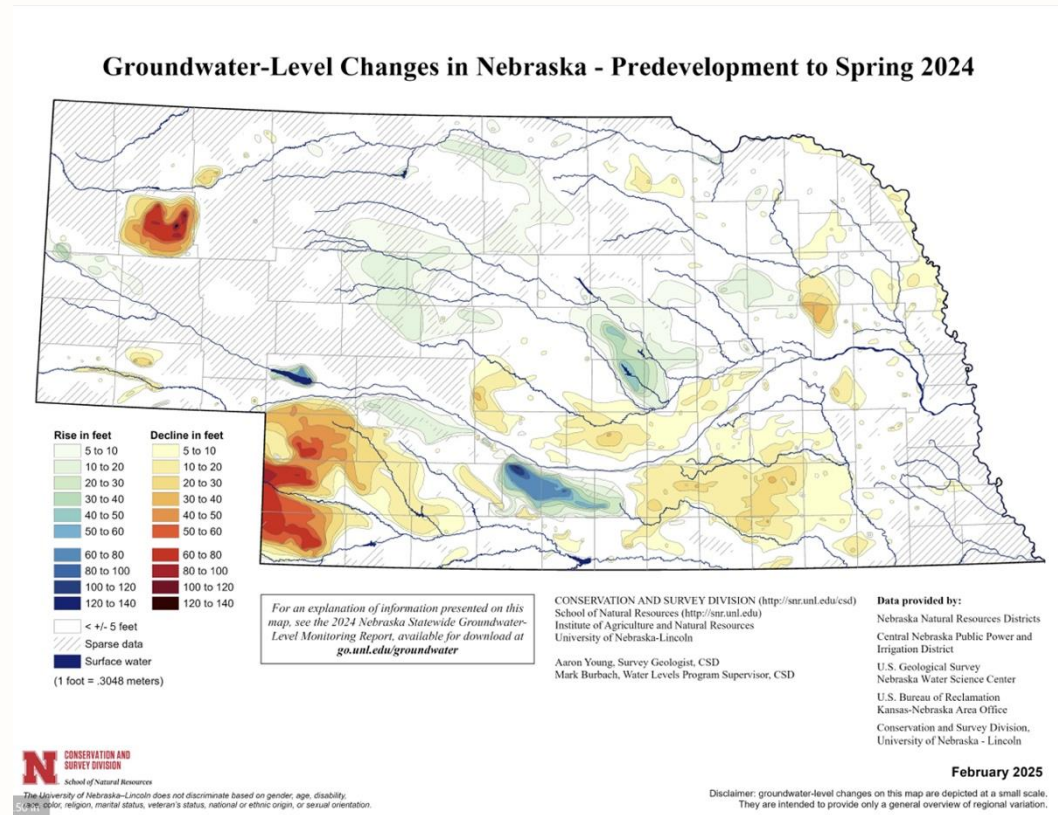


Cultivate ACCESS to Data Science: Equipping high school students with agriculture and water management concepts



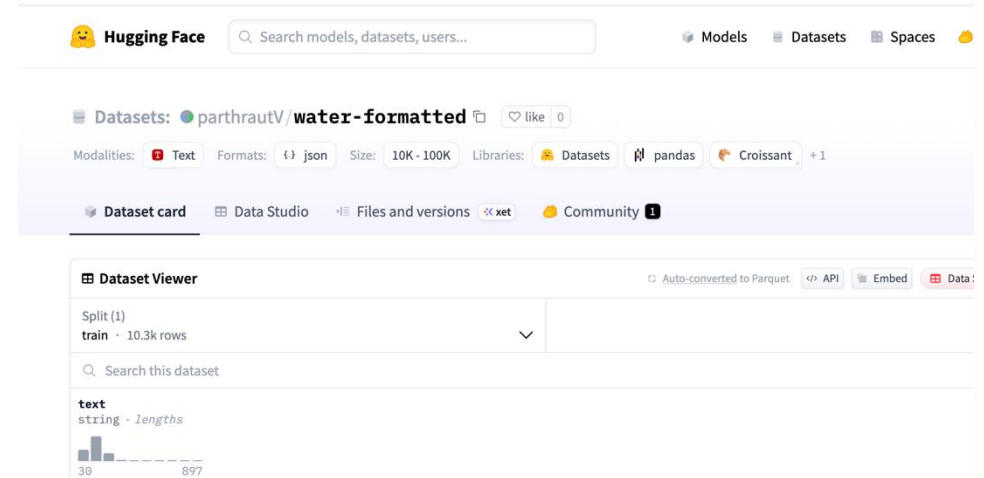
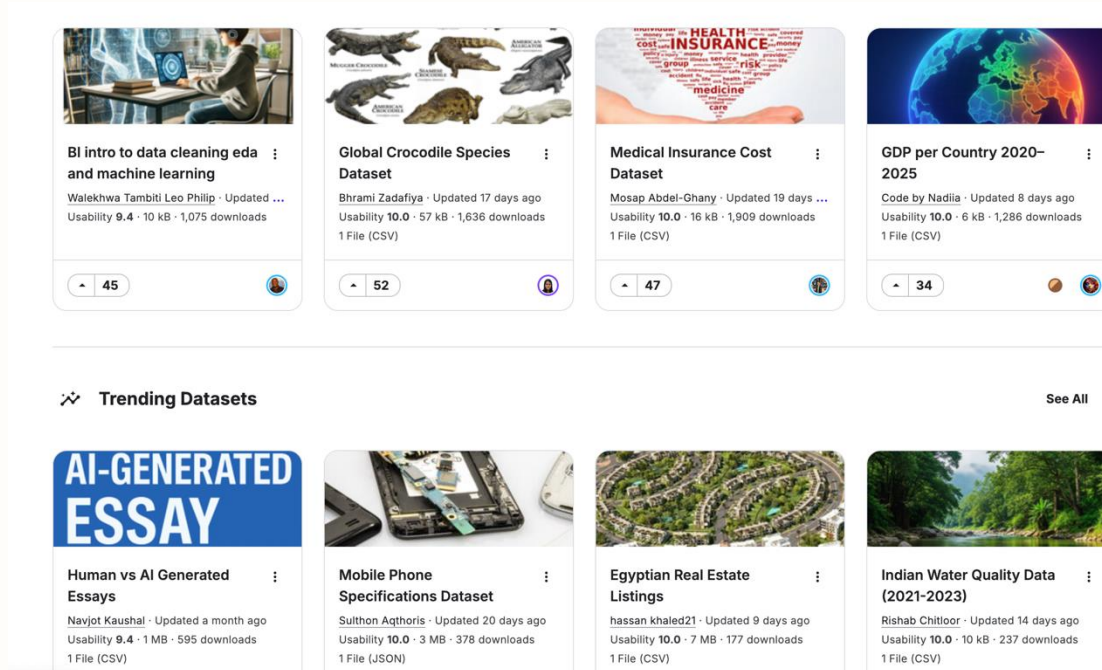
Presented by: Karisma Yumnam

Groundwater in Nebraska



Can we cultivate a next generation of youth capable of thinking critically and responsibly?

What is data science in big data era?



- ❑ Open-source data available from different sources!
- ❑ Computational and cloud infrastructure improved drastically
- ❑ Data Science: Convert data into useful information!

What is data science in big data era?

- ❑ Needs communicating the results clearly
- ❑ Can be overwhelming to high schoolers
- ❑ Need to cultivate curiosity and data literacy in the young students.

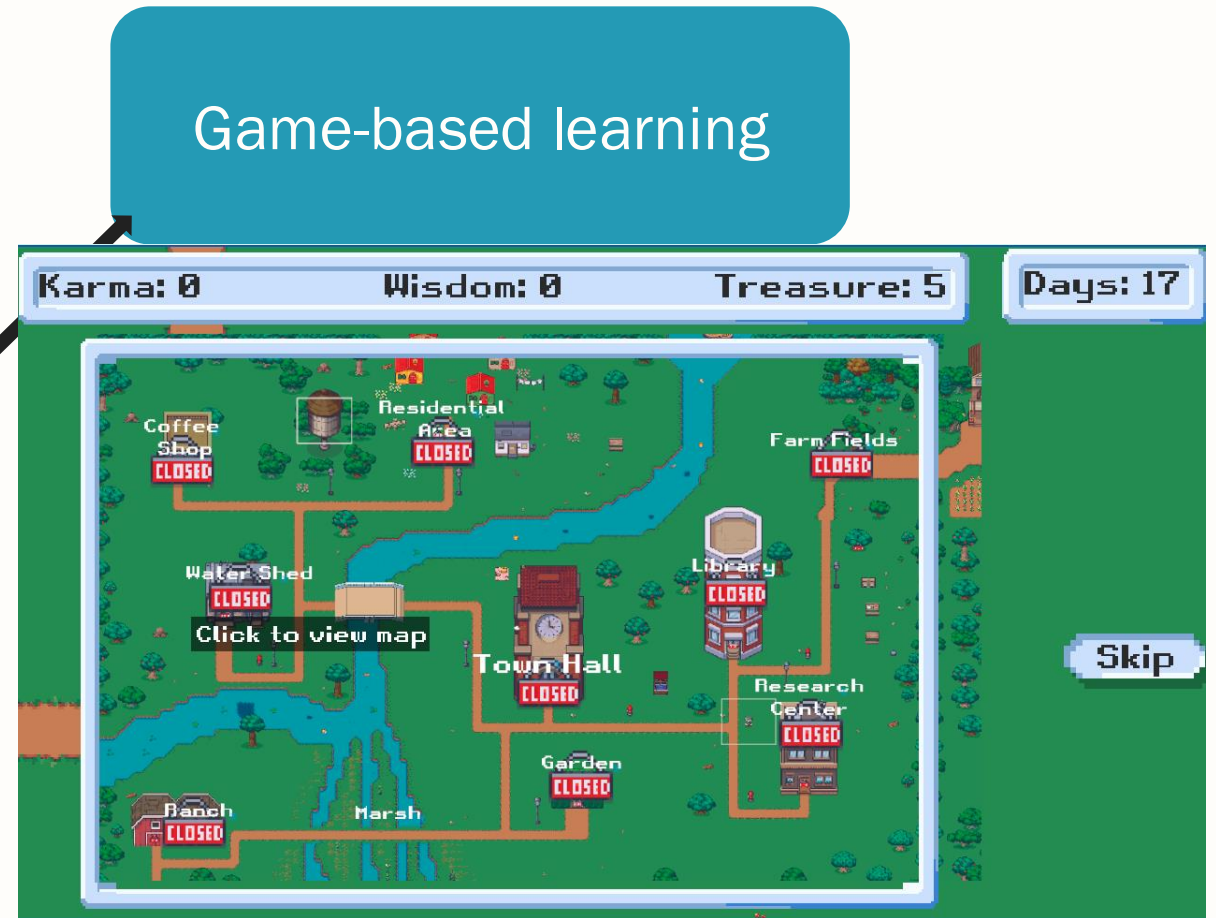
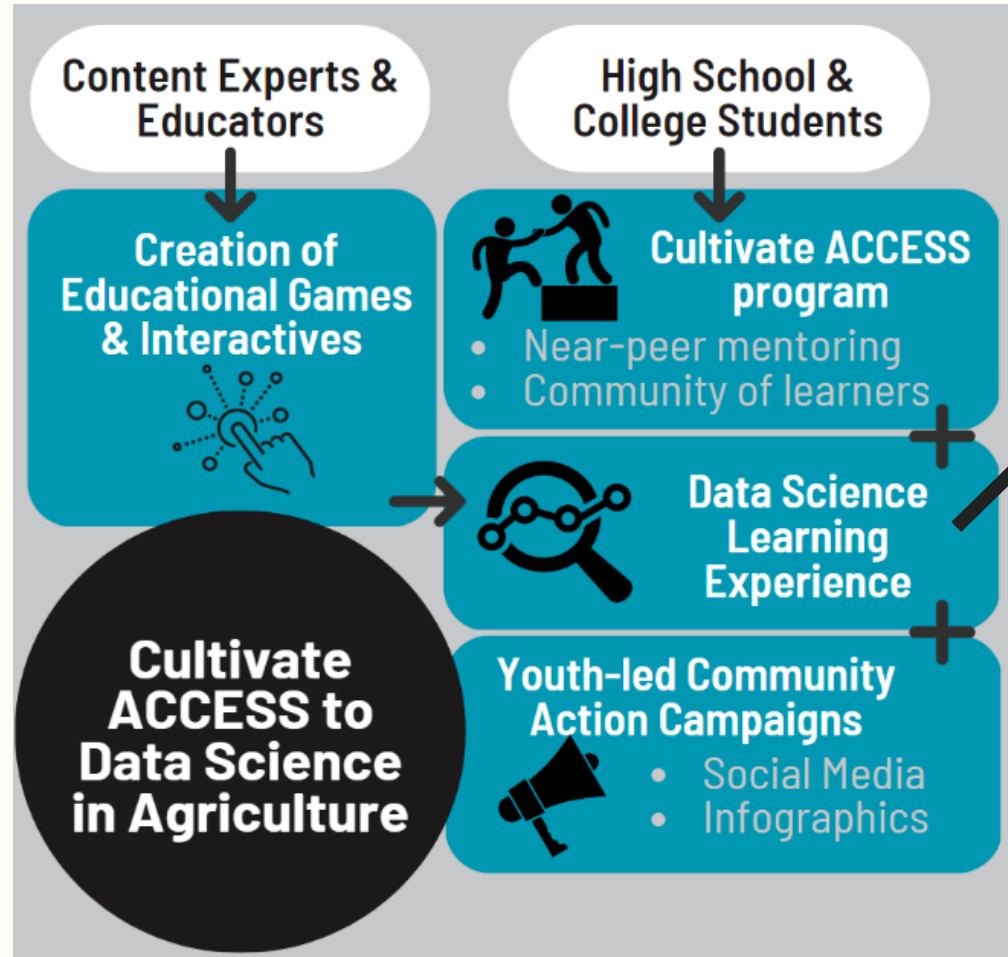


Source: Google

What is system thinking?



Program Overview



Students' Community Action Projects



What is the Missouri river?

- The Missouri River is river that reaches over Kansas, Montana, Iowa, Nebraska South Dakota and North Dakota
- The Missouri has a total course of 2,315



Why is the river receding and why is it such bad thing?

- Melting Snowpacks & Glaciers: In some areas, rivers rely on snowpack or glacier meltwater. River flow could drop as a result of glaciers melting more quickly and snowpacks not refilling as quickly as they once did due to climate change-induced increases in global temperatures.



Water level

- When water recedes this can impact water flow, food production, human health and more
- Droughts can lead to economy loss and more.
- When erosion starts to form you'll see sand and that means the water level has been going down in order for sand banks to be exposed.

Students' Community Action Projects

HOW DID I COME UP WITH THIS?

while thinking about water in Nebraska, I remembered a childhood memory where my family and I would play water balloon fights and after we would drink the leftover water from the hose. My cousins always seemed to get sick after the fights and I wondered if there was any correlation between drinking the hose water and the sickness. Turns out there was and now I am informing you about the harm that a small sip can do.

Why?

you are likely wondering "what's the worst that can happen"? Well after researching and testing water from my own hoses, I found that hose water contains lead, copper, many more harmful chemicals that can cause vomiting, nausea, bladder problems, and damages to the kidney, liver and blood system.

If those reasons aren't bad enough then think about this. Generally hose water, like tap, comes from either the Missouri or Platte rivers. These water sources are most likely contaminated from sewage and well as runoff. Meaning if you drink this water then you are drinking farm chemicals and waste.

Testing results

As you can see, hose water is significantly much worse than the rest. The tap water is a bit better showing you that there may be a small filter on or it's coming from groundwater.

Chemicals	Hose water	Sink water	bottled water	filtered water
total Hardness	50ppm	50ppm	0ppm	0ppm
free chlorine (bromine)	0.5ppm	0.5ppm	0ppm	0ppm
iron	0ppm	0ppm	0ppm	0ppm
Mercury	0ppm	0ppm	0ppm	0ppm
total chlorine	0ppm	0.5ppm	0.5ppm	0.5ppm
copper	3ppm	0.5ppm	0ppm	0ppm
lead	20ppm	0ppm	0ppm	0ppm
zinc	0ppm	2ppm	2ppm	0ppm
Manganese	0.1ppm	0ppm	0.5ppm	0ppm
QAC/Quat	20ppm	10ppm	5ppm	5ppm
Fluoride	10ppm	0ppm	0ppm	0ppm
sodium Chloride	10ppm	500ppm	0ppm	250ppm
hydrogen sulfite	0ppm	0ppm	0ppm	0ppm
total Alkalinity	240ppm	180ppm	40ppm	180ppm
Carbonate	240ppm	240ppm	40ppm	240ppm
PH	8.6ppm	8.6ppm	6.4ppm	6.2ppm

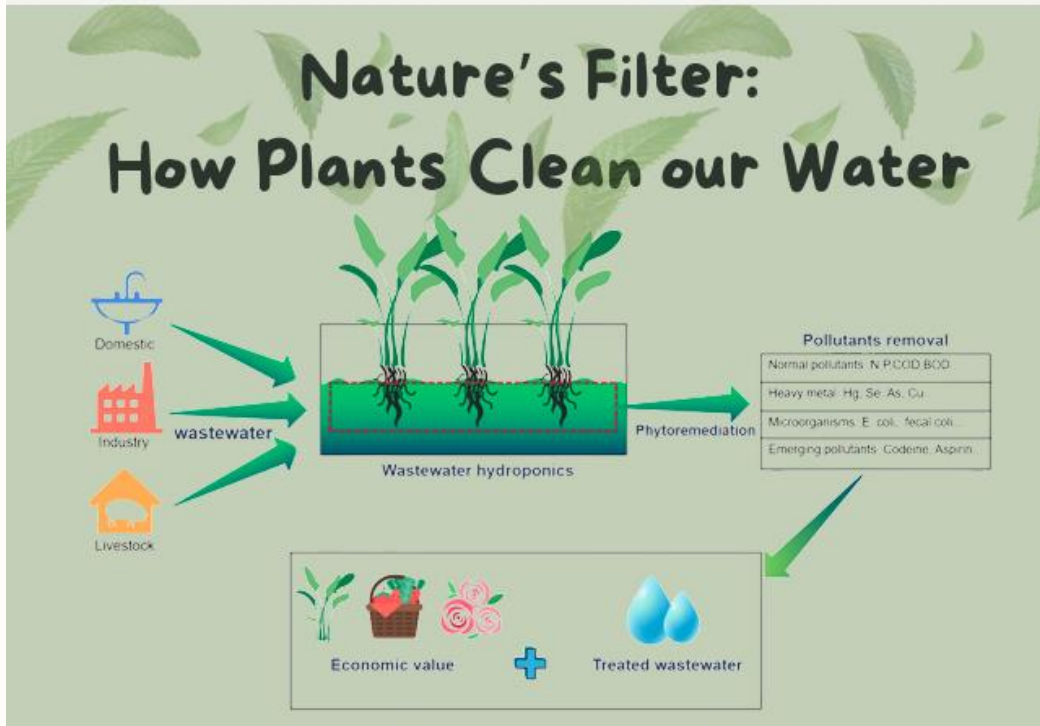
Green- normal
yellow- slightly below or above
red- way above or below average

Changing your ways

By testing out different waters I found the best way to drink water safely is filtered. It may be a bit expensive but in the long run it will be worth it by how healthy it is compared to hose water.



Students' Community Action Projects



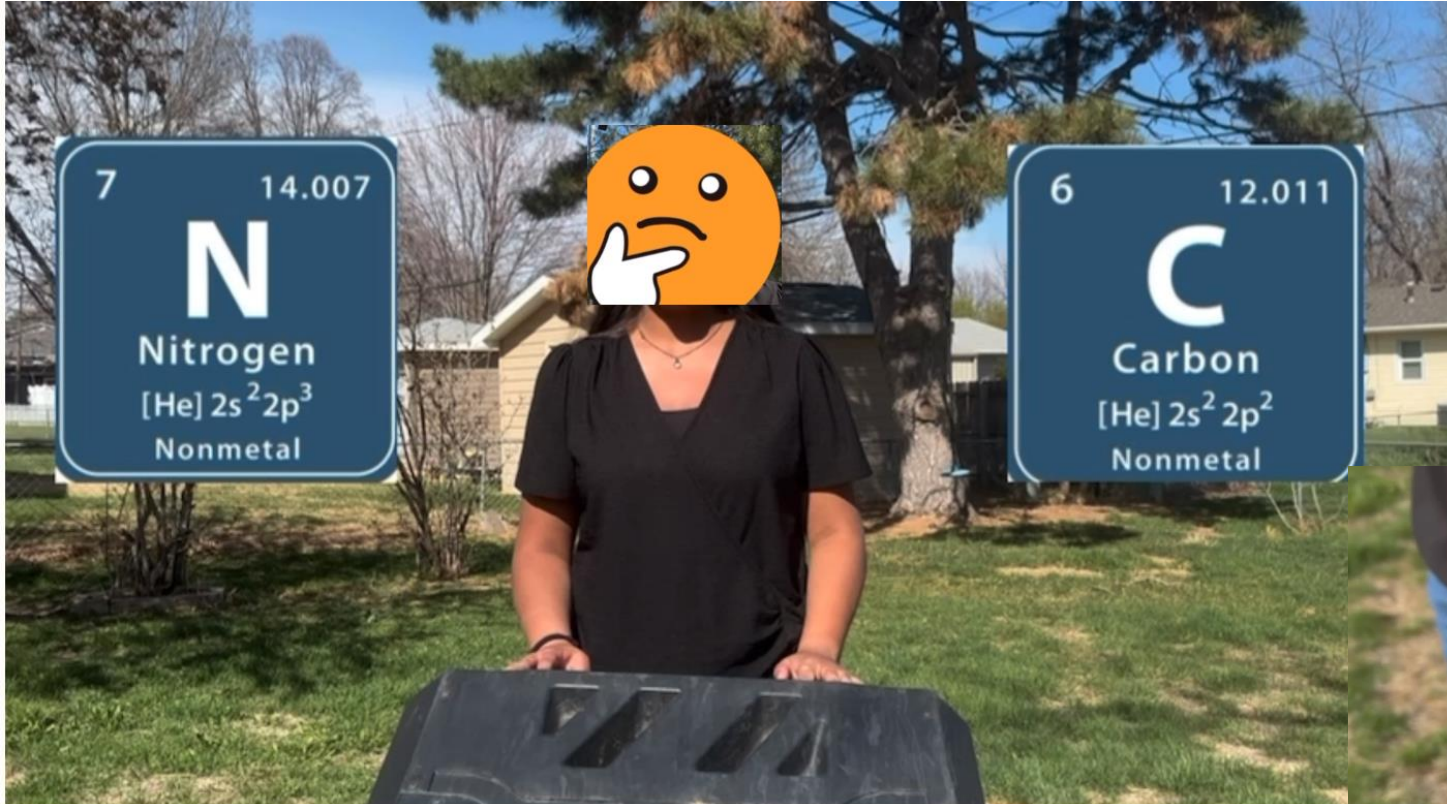
- Plants absorb water through their roots, taking up nutrients and pollutants.
- The roots of plants, slow down water flow, causing suspended particles to settle out of the water.
- Some plants are particularly adept at absorbing heavy metals or harmful chemicals, storing them in their tissues.
- Plants create a habitat for beneficial microbes around their root zones.



UNIVERSITY of NEBRASKA
LINCOLN

Go to the library to see the display of the water irrigation of native plants in Nebraska! Also fill out the Google form

Students' Community Action Projects



Composting
kitchen waste



Cultivate ACCESS community



Thank you for listening!
Any Questions??



CULTIVATE ACCESS
Data Science in Agriculture

